



Computational Materials Science Workshop

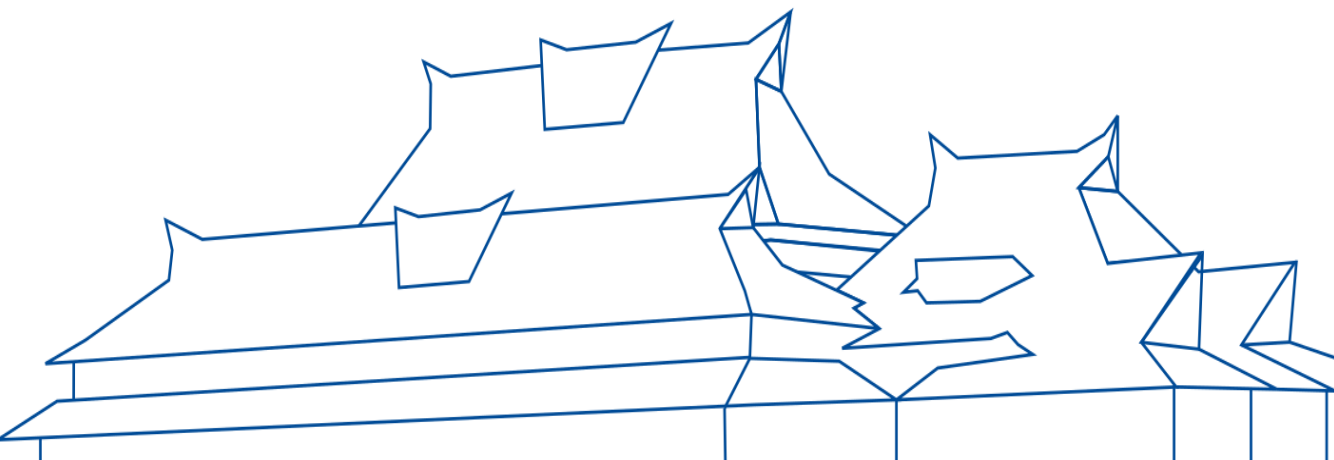
October 7th, 2026

Hands-On Workshop QE: Surface dan Adsorption

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QUANTUM[®]
ESPRESSO



```
&CONTROL
  calculation = 'vc-relax'
  restart_mode = 'from_scratch'
  pseudo_dir = ' /home/u0000/qe_pot/'
  prefix = 'si'
  outdir = './tmp'
  forc_conv_thr = 1.0e-3   #dalam a.u.
/

&SYSTEM
 ibrav = 2
  celldm(1) = 10.25      #dalam Bohr
  nat = 2
  ntyp= 1
  ecutwfc = 60.0, ecutrho = 720.0  #dalam Ry
  nspin = 1
  nbnd = 8
/

&ELECTRONS
  electron_maxstep = 150
  conv_thr = 1.0d-8
/

&IONS
  ion_dynamics = 'bfgs'
/

&CELL
  cell_dynamics = 'bfgs'
  cell_dofree = 'all'
/

ATOMIC_SPECIES
Si 28.086 Si.pbe-n-kjpaw_psl.1.0.0.UPF #pastikan nama file potensial sama

ATOMIC_POSITIONS crystal
Si 0.00 0.00 0.00
Si 0.25 0.25 0.25

K_POINTS automatic
5 5 5 0 0 0
```

&CONTROL : Mengatur jenis dan alur perhitungan

Parameter	Fungsi
calculation	Jenis perhitungan : scf, nscf, bands, relax, vc-relax
restart mode	from_scratch (default) atau restart
pseudo_dir	Folder tempat file pseudopotential .UPF
prefix	Awalan nama file output
outdir	Folder penyimpanan file sementara dan hasil (charge density, wavefunction)
forc_conv_thr	Kriteria konvergensi gaya untuk relaksai (Ry/bohr)
nstep	Jumlah maksimum langkah ionik

```
&CONTROL
  calculation = 'vc-relax'
  restart_mode = 'from_scratch'
  pseudo_dir = ' /home/u0000/qe_pot/'
  prefix = 'si'
  outdir = './tmp'
  forc_conv_thr = 1.0e-3   #dalam a.u.
/
&SYSTEM
  ibrav = 2
  celldm(1) = 10.25        #dalam Bohr
  nat = 2
  ntyp= 1
  ecutwfc = 60.0, ecutrho = 720.0  #dalam Ry
  nspin = 1
  nbnd = 8
/
&ELECTRONS
  electron_maxstep = 150
  conv_thr = 1.0d-8
/
&IONS
  ion_dynamics = 'bfgs'
/
&CELL
  cell_dynamics = 'bfgs'
  cell_dofree = 'all'
/
ATOMIC_SPECIES
Si 28.086 Si.pbe-n-kjpaw_psl.1.0.0.UPF #pastikan nama file potensial sama

ATOMIC_POSITIONS crystal
Si 0.00 0.00 0.00
Si 0.25 0.25 0.25

K_POINTS automatic
5 5 5 0 0 0
```

&SYSTEM : Mendefinisikan sistem fisik

Parameter	Keterangan
ibrav	Tipe kisi bravais (0 = free structure, 1 = simple cubic, 2 = FCC, 3 = BCC, dll)
celldm(1)	Parameter kisi dalam Bohr (0.53 Å)
nat	Jumlah atom dalam sel
ntyp	Jumlah tipe atom
ecutwfc	Cut-off energi gelombang bidang dalam Ry (13.6 eV)
ecutrho	Cut-off energi charge density dan potensial dalam Ry (13.6 eV)
nspin	Pilihan polarized calculation atau bukan (1 = non pc, 2 = pc colinear, dll)
nbnd	Jumlah electronic state (bands) (setengah jumlah elektron (insulator), >= 4 (metal))

```
&CONTROL
  calculation = 'vc-relax'
  restart_mode = 'from_scratch'
  pseudo_dir = ' /home/u0000/qe_pot/'
  prefix = 'si'
  outdir = './tmp'
  forc_conv_thr = 1.0e-3  #dalam a.u.
/
&SYSTEM
 ibrav = 2
  celldm(1) = 10.25      #dalam Bohr
  nat = 2
  ntyp= 1
  ecutwfc = 60.0, ecutrho = 720.0  #dalam Ry
  nspin = 1
  nbnd = 8
/
&ELECTRONS
  electron_maxstep = 150
  conv_thr = 1.0d-8
/
&IONS
  ion_dynamics = 'bfgs'
/
&CELL
  cell_dynamics = 'bfgs'
  cell_dofree = 'all'
/
ATOMIC_SPECIES
Si 28.086 Si.pbe-n-kjpaw_psl.1.0.0.UPF #pastikan nama file potensial sama

ATOMIC_POSITIONS crystal
Si 0.00 0.00 0.00
Si 0.25 0.25 0.25

K_POINTS automatic
5 5 5 0 0 0
```

&ELECTRONS : Konvergensi siklus SCF

Parameter	Keterangan
electron_maxstep	Jumlah iterasi maksimum pada iterasi scf (default = 100)
conv_thr	Kriteria konvergensi scf (Ry)

&IONS : Dinamika ion (untuk relax, vc-relax, md)

Parameter	Keterangan
ion_dynamics	'bfgs' (default untuk relax), Quasi-Newton 'damp', Quick-Min Verlet

&CELL : Dinamika sel (untuk vc-relax dan vc-md)

Parameter	Keterangan
cell_dynamics	'none', no dynamics 'sd', steepest descent 'bfgs', Quasi-Newton
cell_dofree	Derajat kebebasan sel ('all', 'x', 'xy', '2Dxy', 'ibrav', dll)

```
&CONTROL
  calculation = 'vc-relax'
  restart_mode = 'from_scratch'
  pseudo_dir = ' /home/u0000/qe_pot/'
  prefix = 'si'
  outdir = './tmp'
  forc_conv_thr = 1.0e-3  #dalam a.u.
/
&SYSTEM
 ibrav = 2
  celldm(1) = 10.25      #dalam Bohr
  nat = 2
  ntyp= 1
  ecutwfc = 60.0, ecutrho = 720.0  #dalam Ry
  nspin = 1
  nbnd = 8
/
&ELECTRONS
  electron_maxstep = 150
  conv_thr = 1.0d-8
/
&IONS
  ion_dynamics = 'bfgs'
/
&CELL
  cell_dynamics = 'bfgs'
  cell_dofree = 'all'
/
```

```
ATOMIC_SPECIES
Si 28.086 Si.pbe-n-kjpaw_psl.1.0.0.UPF #pastikan nama file potensial sama

ATOMIC_POSITIONS crystal
Si 0.00 0.00 0.00
Si 0.25 0.25 0.25

K_POINTS automatic
5 5 5 0 0 0
```

ATOMIC POSITIONS

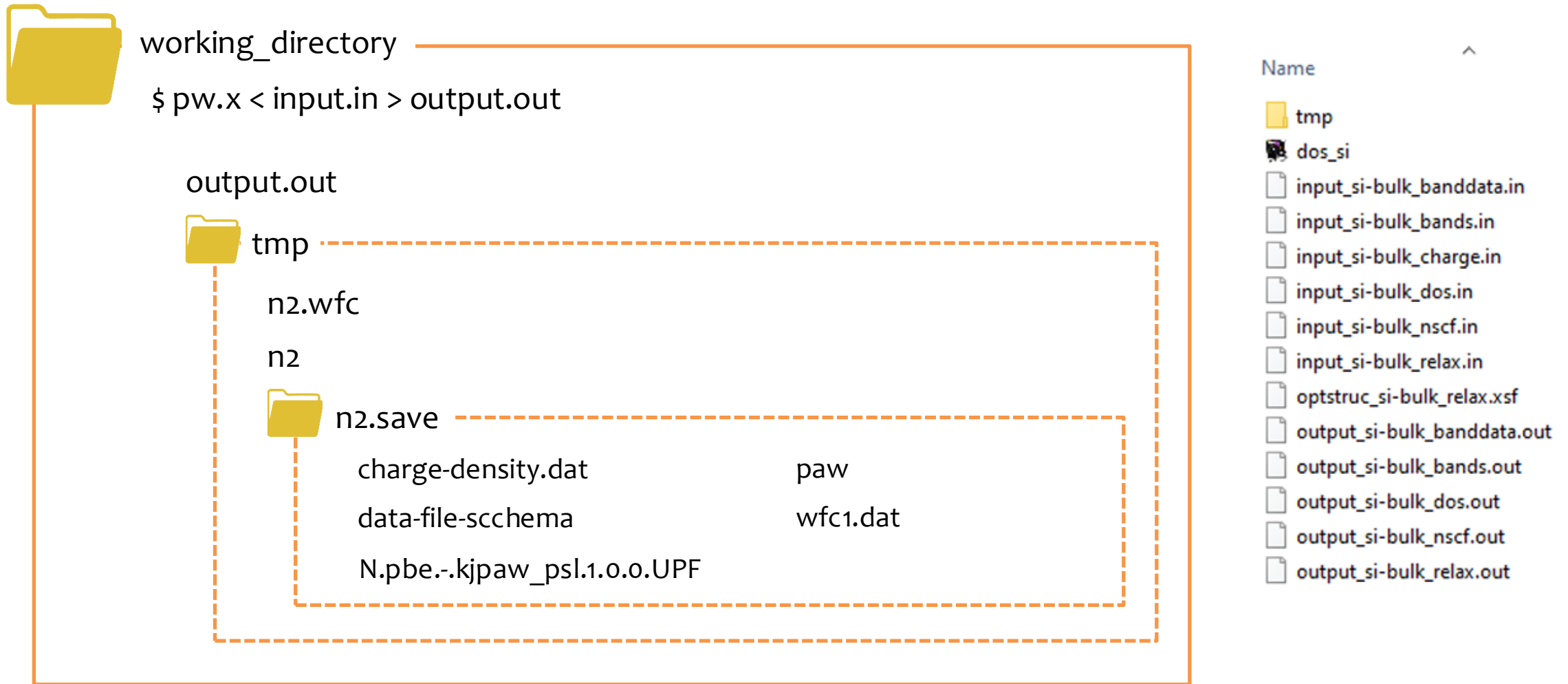
Parameter	Keterangan
alat	Koordinat kartesian dalam unit parameter kisi a atau celldm(1)
bohr	Koordinat kartesian dalam unit bohr
angstrom	Koordinat kartesian dalam unit Angstrom
crystal	Koordinat yang sesuai dengan CELL_PARAMETERS (fractional)

KPOINTS

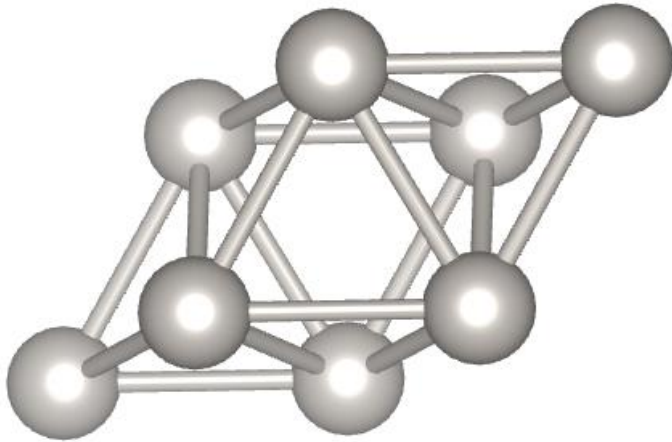
Parameter	Keterangan
automatic	Monkhorst-Pack
gamma	Gamma point only

CELL_PARAMETERS (jika ibrav = 0)

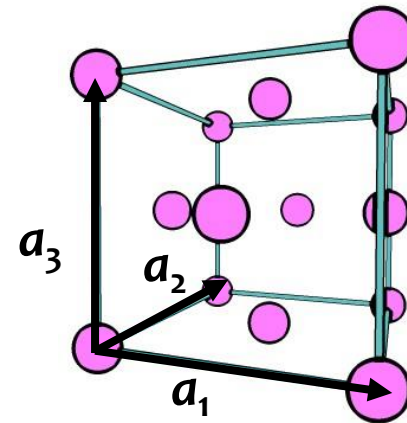
Parameter	Keterangan
alat	Koordinat kartesian dalam unit parameter kisi a atau celldm(1)
bohr	Koordinat kartesian dalam unit bohr



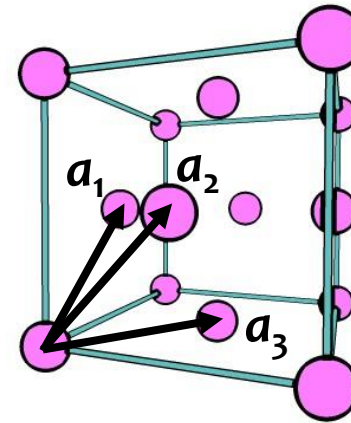
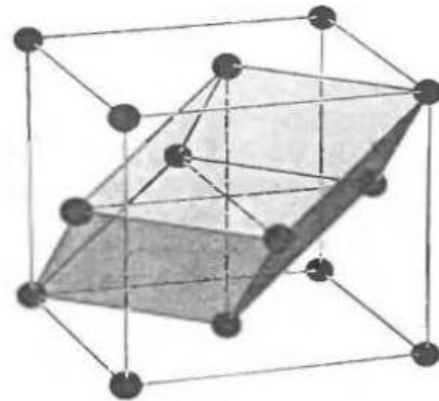
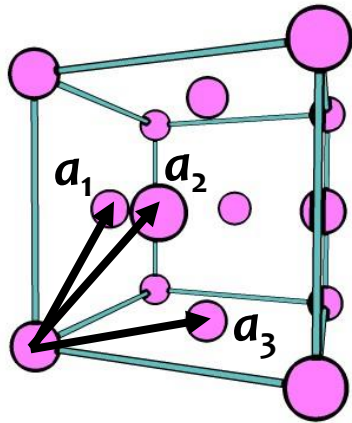
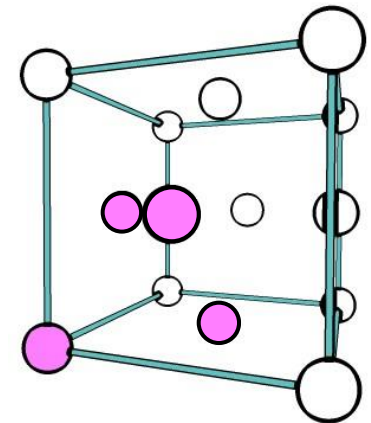
Folder tmp mengandung semua file yang dihasilkan pw.x
File-file ini penting untuk “post_processing” seperti DOS, Band structure, Charge density



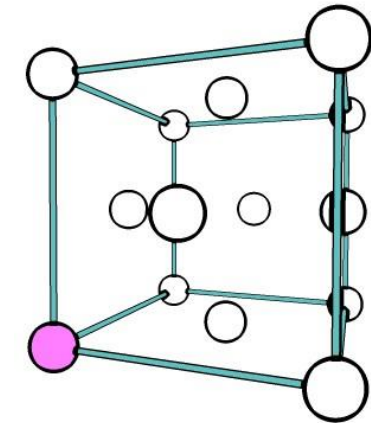
FCC Pt using primitive lattice vectors



$$a_1 = a x, a_2 = a y, a_3 = a z$$



$$a_1 = \frac{a}{2}(y + z), a_2 = \frac{a}{2}(x + z), a_3 = \frac{a}{2}(x + y)$$



- Pt: cubic FCC lattice
- Optimasi sel
- Lattice constant = $3.92 \text{ \AA} = 7.407726 \text{ Bohr}$
- Primitive unit cell $\rightarrow$ 1 Pt atom per unit cell
- Core electrons: PAW (Projector Augmented Wave)
- Fungsional PBE untuk Exchange-Correlation Functional
- $9 \times 9 \times 9$ k-points grid untuk integrasi pada grid BZ
- Ecut-off = 47 Ry (= 640 eV)
- Smearing = Methfessel-Paxton

```
&CONTROL
  calculation='vc-relax',
  restart_mode='from_scratch'
  pseudo_dir='./', outdir='./tmp/'
  nstep=100, forc_conv_thr=0.0005
/
&SYSTEM
  ibrav=2,
  celldm(1)=7.407726,
  nat=1, ntyp=1,
  ecutwfc=47, ecutrho=400,
  occupations='smearing', smearing='mp',
  degauss=0.005,
/
&ELECTRONS
  electron_maxstep = 200
  conv_thr = 1.0d-6
/
&IONS
  ion_dynamics = 'bfgs'
/
&CELL
  cell_dynamics = 'bfgs'
  cell_dofree = 'all'
/
ATOMIC_SPECIES
  Pt 195.084 Pt.pbe-n-kjpaw_psl.1.0.0.UPF
```

```
ATOMIC_POSITIONS {alat}
Pt 0.000000000d0 0.000000000d0 0.000000000d0

K_POINTS {automatic}
9 9 9 0 0 0
```

the Fermi energy is 16.7646 ev

! total energy = -747.30005887 Ry

total all-electron energy = -36827.487966 Ry
estimated scf accuracy < 3.9E-10 Ry
smearing contrib. (-TS) = 0.00007362 Ry
internal energy E=F+TS = -747.30013249 Ry

The total energy is F=E-TS. E is the sum of the following terms:

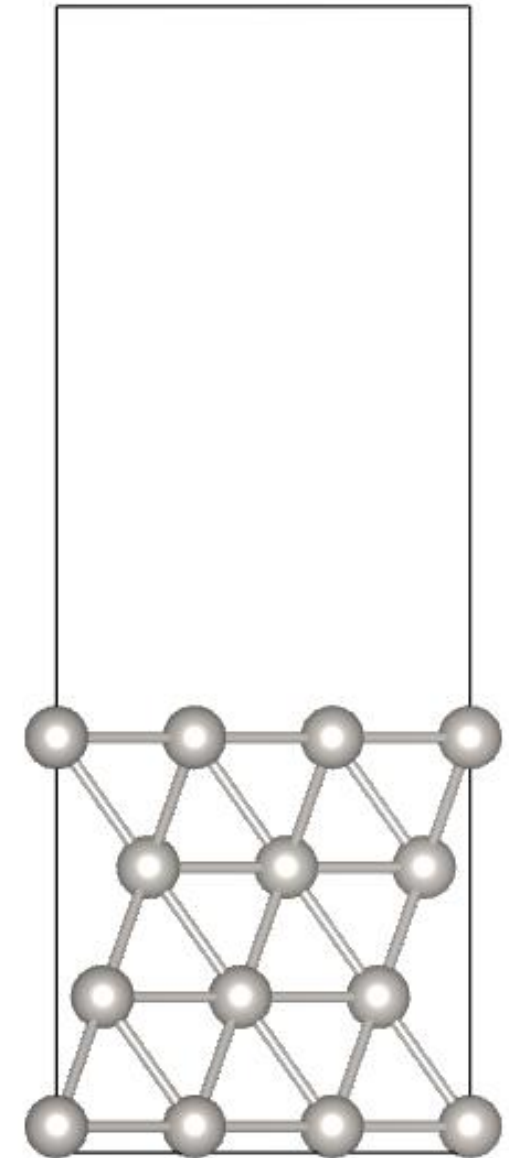
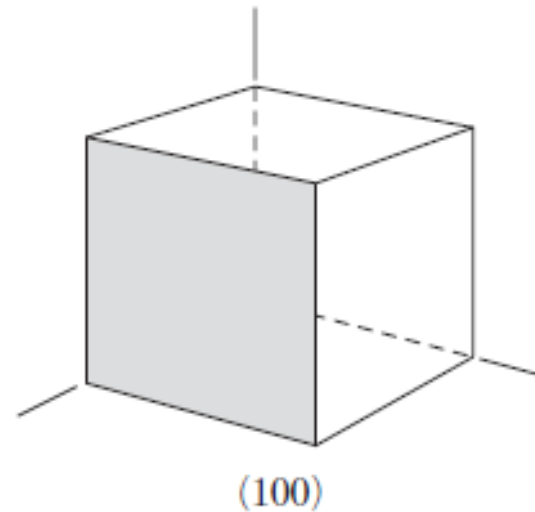
one-electron contribution = 17.98470974 Ry
hartree contribution = 4.39940574 Ry
xc contribution = -23.88931399 Ry
ewald contribution = -61.07676143 Ry
one-center paw contrib. = -684.71817255 Ry

convergence has been achieved in 11 iterations

grep ! pt_bulk.out

```
haris@yamanote:~/test-opt/qe$ grep ! pt_bulk.out
! total energy = -747.29851836 Ry
! total energy = -747.29972509 Ry
! total energy = -747.30001591 Ry
! total energy = -747.30002598 Ry
! total energy = -747.30002670 Ry
! total energy = -747.30002657 Ry
! total energy = -747.30002672 Ry
! total energy = -747.30002673 Ry
! total energy = -747.30005887 Ry
```

- Sebagian parameter sama dengan bulk
 - EXCEPT: Kalkulasi relaksasi posisi atom saja
- Penggunaan super-cell, definisi slab: 4 layer
- Pt(100), 5 atom dalam unit cell
- Vacuum space
-
- $3 \times 3 \times 1$. Pada arah-z rentang periodic sangat besar sehingga BZ pada arah-z sangat kecil, cukup 1 k-point
 - 4 layer Pt(100), 3 layer fixed, 1 layers relax in 3 directions



```

&CONTROL
  calculation='relax',
  restart_mode='from_scratch'
  pseudo_dir='./', outdir='./tmp/'
  nstep=100
  forc_conv_thr=0.001
/
&SYSTEM
  ibrav=0,
  nat=36, ntyp=1,
  ecutwfc=47, ecutrho=400,
  occupations='smearing', smearing='mp',
  degauss=0.005,
/
&ELECTRONS
  electron_maxstep = 200
  conv_thr = 1.0d-6
/
&IONS
  ion_dynamics = 'bfgs'
/
ATOMIC_SPECIES
  Pt 195.084 Pt.pbe-n-kjpaw_psl.1.0.0.UPF

```

```

ATOMIC_POSITIONS {crystal}
  Pt 0.0000000000d0 0.0000000000d0 0.0226286200d0 0 0 0
  Pt 0.0000000816d0 0.3333333077d0 0.0226286200d0 0 0 0
  Pt 0.0000001633d0 0.6666666155d0 0.0226286200d0 0 0 0
  Pt 0.3333331445d0 0.0000000000d0 0.0226286200d0 0 0 0
  Pt 0.3333332261d0 0.3333333077d0 0.0226286200d0 0 0 0
  Pt 0.3333333077d0 0.6666666155d0 0.0226286200d0 0 0 0
  Pt 0.6666662889d0 0.0000000000d0 0.0226286200d0 0 0 0
  Pt 0.6666663706d0 0.3333333077d0 0.0226286200d0 0 0 0
  Pt 0.6666664522d0 0.6666666155d0 0.0226286200d0 0 0 0
  Pt 0.2222221236d0 0.1111111026d0 0.1357924578d0 0 0 0
  Pt 0.2222222053d0 0.4444444103d0 0.1357924578d0 0 0 0
  Pt 0.2222222869d0 0.7777777180d0 0.1357924578d0 0 0 0
  Pt 0.5555552681d0 0.1111111026d0 0.1357924578d0 0 0 0
  Pt 0.5555553497d0 0.4444444103d0 0.1357924578d0 0 0 0
  Pt 0.5555554314d0 0.7777777180d0 0.1357924578d0 0 0 0
  Pt 0.8888884126d0 0.1111111026d0 0.1357924578d0 0 0 0
  Pt 0.8888884942d0 0.4444444103d0 0.1357924578d0 0 0 0
  Pt 0.8888885758d0 0.7777777180d0 0.1357924578d0 0 0 0
  Pt 0.1111111026d0 0.2222222053d0 0.2489531043d0 0 0 0
  Pt 0.1111111842d0 0.5555555130d0 0.2489531043d0 0 0 0
  Pt 0.1111112658d0 0.8888888207d0 0.2489531043d0 0 0 0
  Pt 0.4444442470d0 0.2222222053d0 0.2489531043d0 0 0 0
  Pt 0.4444443287d0 0.5555555130d0 0.2489531043d0 0 0 0
  Pt 0.4444444103d0 0.8888888207d0 0.2489531043d0 0 0 0
  Pt 0.7777773915d0 0.2222222053d0 0.2489531043d0 0 0 0
  Pt 0.7777774731d0 0.5555555130d0 0.2489531043d0 0 0 0
  Pt 0.7777775548d0 0.8888888207d0 0.2489531043d0 0 0 0

```

```

Pt      0.0000000000d0    0.0000000000d0    0.3621137508d0
Pt      0.0000000816d0    0.3333333077d0    0.3621137508d0
Pt      0.0000001633d0    0.6666666155d0    0.3621137508d0
Pt      0.3333331445d0    0.0000000000d0    0.3621137508d0
Pt      0.3333332261d0    0.3333333077d0    0.3621137508d0
Pt      0.3333333077d0    0.6666666155d0    0.3621137508d0
Pt      0.6666662889d0    0.0000000000d0    0.3621137508d0
Pt      0.6666663706d0    0.3333333077d0    0.3621137508d0
Pt      0.6666664522d0    0.6666666155d0    0.3621137508d0

```

```

K_POINTS {automatic}
3 3 1 0 0 0

```

```

CELL_PARAMETERS {angstrom}
8.315580000000    0.000000000000    0.000000000000
-4.157790000000    7.201500000000    0.000000000000
0.000000000000    0.000000000000    20.000000000000

```

```

haris@yamanote:~/test-opt/qe$ grep ! Pt100.out

```

```

!      total energy      = -26901.95132542 Ry
!      total energy      = -26901.95464889 Ry
!      total energy      = -26901.95765691 Ry
!      total energy      = -26901.96095110 Ry
!      total energy      = -26901.96331043 Ry
!      total energy      = -26901.96344259 Ry
!      total energy      = -26901.96344319 Ry

```

Energi yang dibutuhkan untuk membuat permukaan atau eksese energi akibat pembentukan permukaan. Selalu endotermik

$$E_{surf} = \frac{1}{2} (E_{slab} - N * E_{bulk})$$

$$E_{slab} = -26901.9634 \text{ Ry}$$

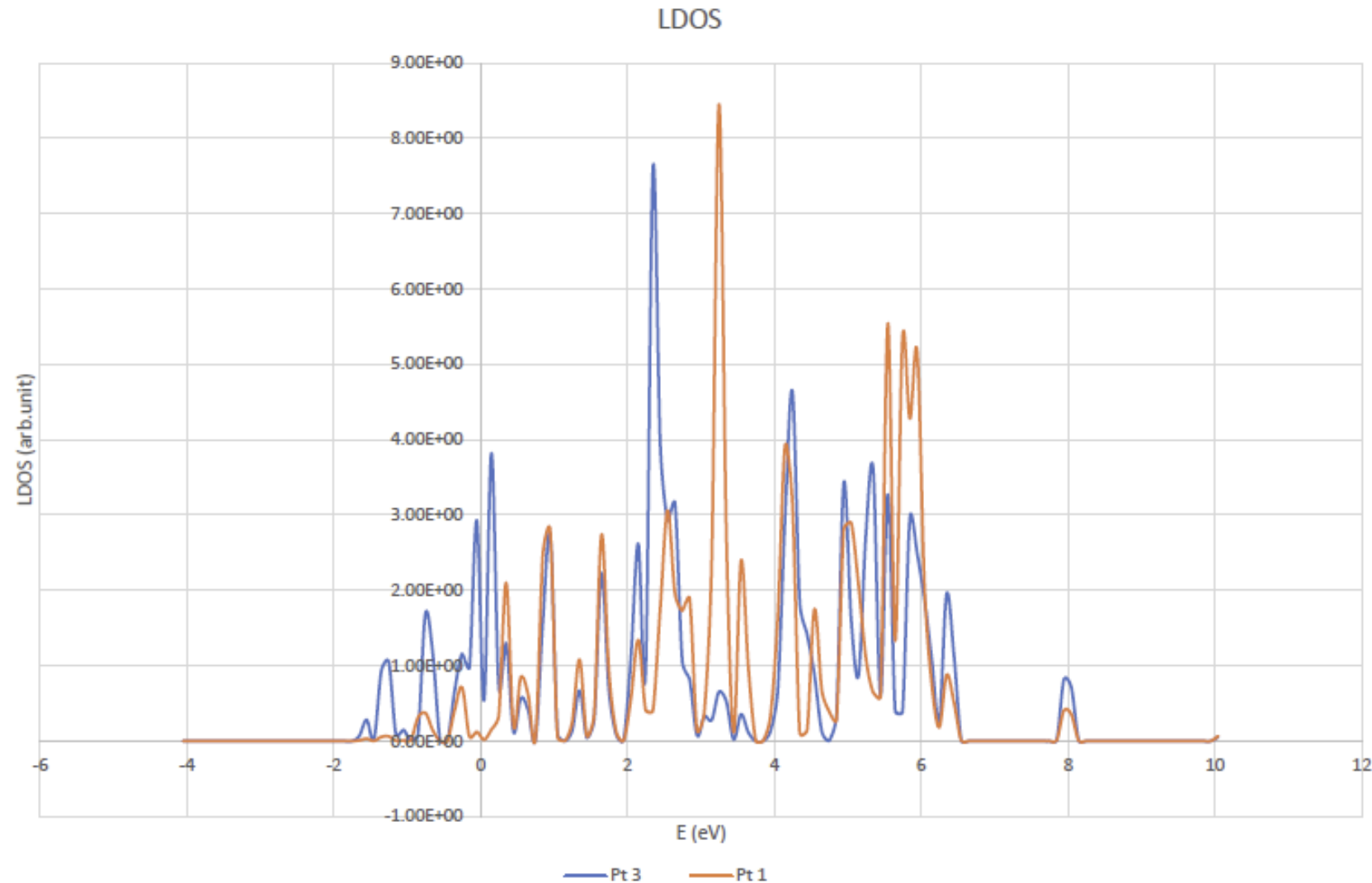
$$E_{bulk} = -747.3000 \text{ Ry}$$

$$N = 36$$

$$E_{surf} = 0.4183 \text{ Ry} = 5.6888 \text{ eV}$$

```
&DOS
  prefix = 'Pt100_slab'
  outdir = './tmp'
  fildos = 'Pt100_slab_dos'
  emin = -10.0
  emax = 10.0
/
```

```
&projwfc
  prefix = 'Pt100_slab'
  outdir = './tmp'
  emin = -10.0
  emax = 10.0
  DeltaE = 0.01
  filpdos = 'Pt100_slab_pdos'
/
```



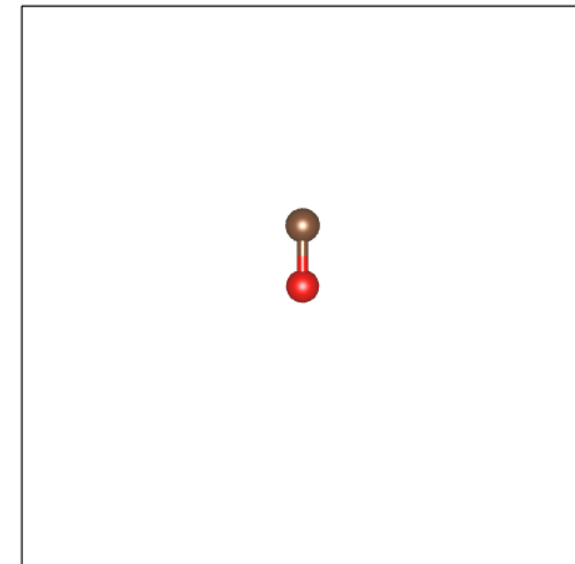
Perbedaan LDOS antara atom bagian dalam (Pt 3) dan permukaan (Pt 1) akibat perbedaan koordinasi dengan lingkungannya.

```
&CONTROL
calculation='relax'
restart_mode='from_scratch'
pseudo_dir='./', outdir='./tmp/'
nstep=100, forc_conv_thr=0.001,
/
&SYSTEM
ibrav=0
celldm(1)=18.8972612457d0,
ntyp=2,nat=2
ecutwfc=47, ecutrho=360
occupations='smearing', smearing = 'gaussian'
degauss = 0.005
/
&ELECTRONS
electron_maxstep=300
conv_thr = 1.0d-6
/
&IONS
ion_dynamics='bfgs'
/
ATOMIC_SPECIES
C 12.0107 C.pbe-n-kjpaw_ps1.1.0.0.UPF
O 15.9999 O.pbe-n-kjpaw_ps1.1.0.0.UPF

ATOMIC_POSITIONS {crystal}
O 0.5000000360d0 0.5000000360d0 0.5000000360d0
C 0.5000000360d0 0.5000000360d0 0.6086350438d0
```

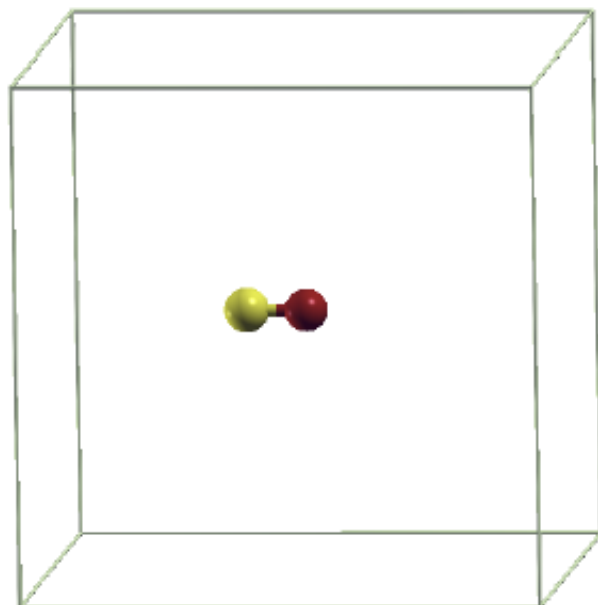
```
K_POINTS {automatic}
1 1 1 0 0 0

CELL_PARAMETERS {alat}
1.0000000000d0 0.0000000000d0 0.0000000000d0
0.0000000000d0 1.0000000000d0 0.0000000000d0
0.0000000000d0 0.0000000000d0 1.0000000000d0
```



Satu molekul CO ditempatkan dalam supercell yang besar untuk meminimalisir interaksi dengan tetangga periodiknya.

Perhatikan nilai parameter celldm(1) yang besar



Forces acting on atoms (cartesian axes, Ry/au):

atom	1	type	2	force =	0.000000000	0.000000000	-0.00037998
atom	2	type	1	force =	0.000000000	0.000000000	0.00037998

Total force = 0.000380 Total SCF correction = 0.000067

SCF correction compared to forces is large: reduce conv_thr to get better values

Energy error = 1.2E-05 Ry

Gradient error = 3.8E-04 Ry/Bohr

bfgs converged in 5 scf cycles and 3 bfgs steps
(criteria: energy < 1.0E-04 Ry, force < 5.0E-04 Ry/Bohr)

End of BFGS Geometry Optimization

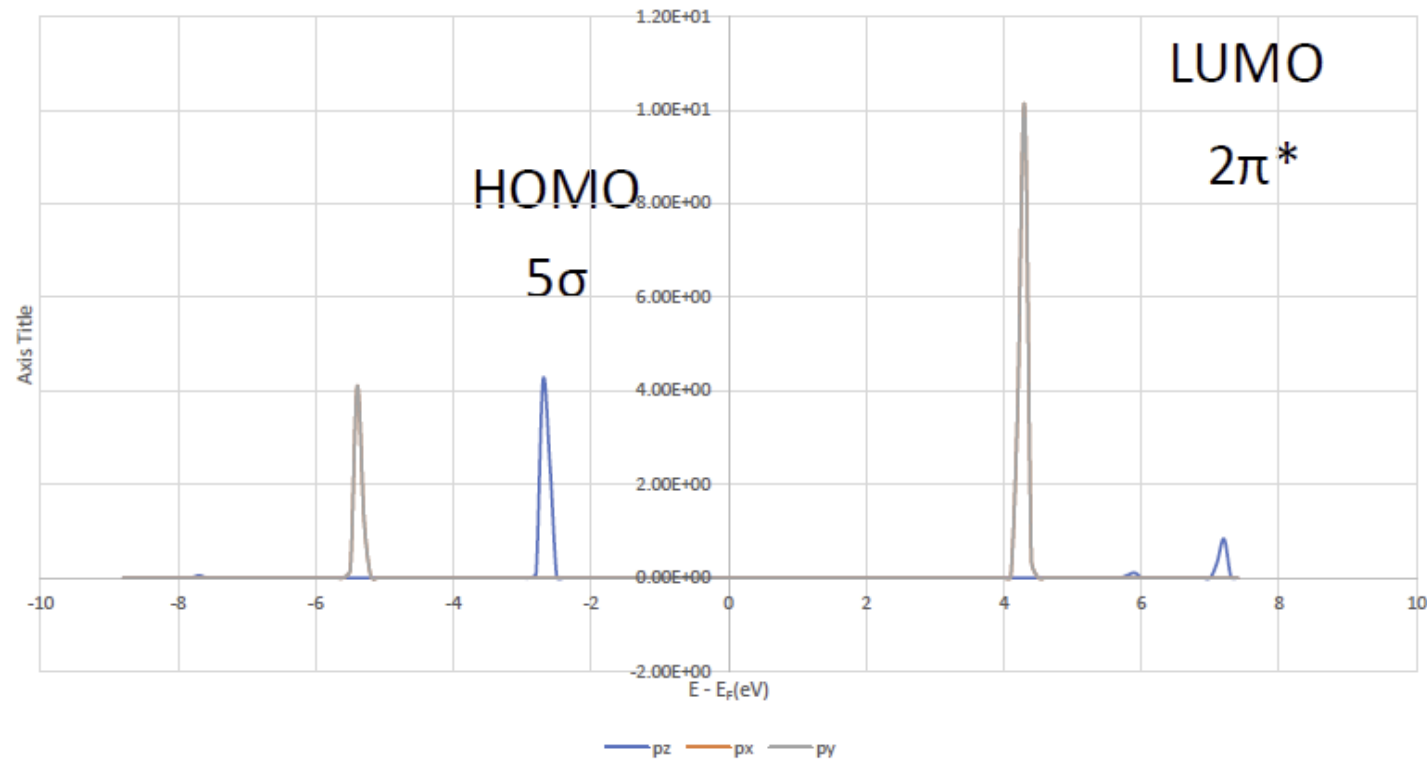
Final energy = -60.0104015632 Ry

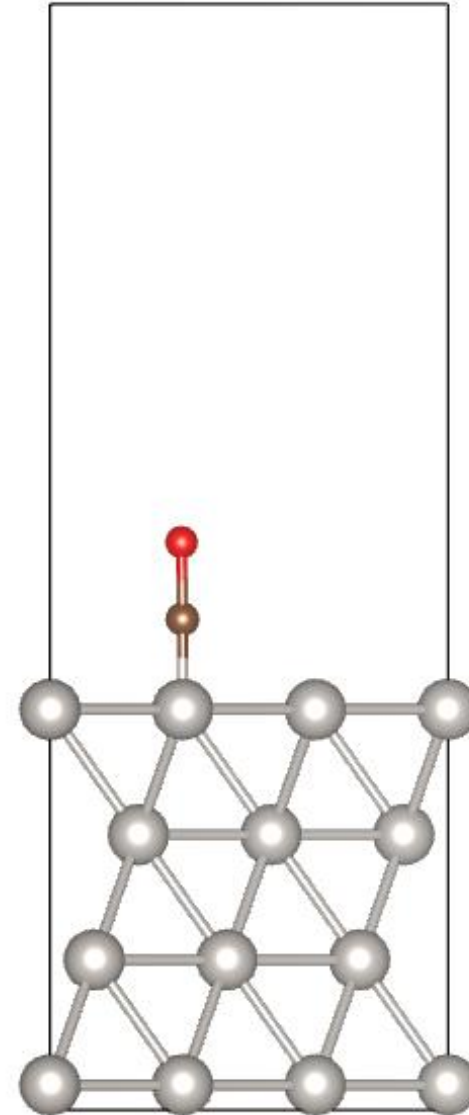
$$E_{\text{mol}} = -60.01040 \text{ Ry}$$

$$d_{\text{C-O}} = 1.1398 \text{ \AA}$$

```
&DOS
  prefix = 'co_isolated'
  outdir = './tmp'
  fildos = 'co_isolated_dos'
  emin = -10.0
  emax = 10.0
/
```

```
&projwfc
  prefix   = 'co_isolated'
  outdir   = './tmp'
  emin     = -10.0
  emax     = 10.0
  DeltaE   = 0.01
  filpdos  = 'co_isolated_pdos'
/
```





```

&CONTROL
  calculation='relax',
  restart_mode='from_scratch'
  pseudo_dir='./', outdir='./tmp/'
  nstep=100
  forc_conv_thr=0.001
/
&SYSTEM
  ibrav=0,
  nat=38, ntyp=3,
  ecutwfc=47, ecutrho=400,
  occupations='smearing',
  smearing='mp',
  degauss=0.005,
/
&ELECTRONS
  electron_maxstep = 200
  conv_thr = 1.0d-6
/
&IONS
  ion_dynamics = 'bfgs'
/
ATOMIC_SPECIES
Pt 195.084 Pt.pbe-n-kjpaw_ps1.1.0.0.UPF
C 12.0107 C.pbe-n-kjpaw_ps1.1.0.0.UPF
O 15.9999 O.pbe-n-kjpaw_ps1.1.0.0.UPF

```

```

ATOMIC_POSITIONS {crystal}
Pt 0.0000000000d0 0.0000000000d0 0.0226286200d0 0 0 0
Pt 0.0000000816d0 0.3333333077d0 0.0226286200d0 0 0 0
Pt 0.0000001633d0 0.6666666155d0 0.0226286200d0 0 0 0
Pt 0.3333331445d0 0.0000000000d0 0.0226286200d0 0 0 0
Pt 0.3333332261d0 0.3333333077d0 0.0226286200d0 0 0 0
Pt 0.3333333077d0 0.6666666155d0 0.0226286200d0 0 0 0
Pt 0.6666662889d0 0.0000000000d0 0.0226286200d0 0 0 0
Pt 0.6666663706d0 0.3333333077d0 0.0226286200d0 0 0 0
Pt 0.6666664522d0 0.6666666155d0 0.0226286200d0 0 0 0
Pt 0.2222221236d0 0.1111111026d0 0.1357924578d0 0 0 0
Pt 0.2222222053d0 0.4444444103d0 0.1357924578d0 0 0 0
Pt 0.2222222869d0 0.7777777180d0 0.1357924578d0 0 0 0
Pt 0.5555552681d0 0.1111111026d0 0.1357924578d0 0 0 0
Pt 0.5555553497d0 0.4444444103d0 0.1357924578d0 0 0 0
Pt 0.5555554314d0 0.7777777180d0 0.1357924578d0 0 0 0
Pt 0.8888884126d0 0.1111111026d0 0.1357924578d0 0 0 0
Pt 0.8888884942d0 0.4444444103d0 0.1357924578d0 0 0 0
Pt 0.8888885758d0 0.7777777180d0 0.1357924578d0 0 0 0
Pt 0.1111111026d0 0.2222222053d0 0.2489531043d0 0 0 0
Pt 0.1111111842d0 0.5555555130d0 0.2489531043d0 0 0 0
Pt 0.1111112658d0 0.8888888207d0 0.2489531043d0 0 0 0
Pt 0.4444442470d0 0.2222222053d0 0.2489531043d0 0 0 0
Pt 0.4444443287d0 0.5555555130d0 0.2489531043d0 0 0 0
Pt 0.4444444103d0 0.8888888207d0 0.2489531043d0 0 0 0
Pt 0.7777773915d0 0.2222222053d0 0.2489531043d0 0 0 0
Pt 0.7777774731d0 0.5555555130d0 0.2489531043d0 0 0 0
Pt 0.7777775548d0 0.8888888207d0 0.2489531043d0 0 0 0

```

```

Pt  0.0000000000d0  0.0000000000d0  0.3621137508d0
Pt  0.0000000816d0  0.3333333077d0  0.3621137508d0
Pt  0.0000001633d0  0.6666666155d0  0.3621137508d0
Pt  0.3333331445d0  0.0000000000d0  0.3621137508d0
Pt  0.3333332261d0  0.3333333077d0  0.3621137508d0
Pt  0.3333333077d0  0.6666666155d0  0.3621137508d0
Pt  0.6666662889d0  0.0000000000d0  0.3621137508d0
Pt  0.6666663706d0  0.3333333077d0  0.3621137508d0
Pt  0.6666664522d0  0.6666666155d0  0.3621137508d0
C   0.3380396924d0  0.3329748621d0  0.4438908195d0
O   0.3425119381d0  0.3304968746d0  0.5133467477d0

```

```

K_POINTS {automatic}
  3 3 1 0 0 0

```

```

CELL_PARAMETERS angstrom
  8.31558000000  0.00000000000  0.00000000000
 -4.15779000000  7.20150000000  0.00000000000
  0.00000000000  0.00000000000  20.00000000000

```

$$E_{ads} = E_{CO/slab} - E_{slab} - E_{CO}$$

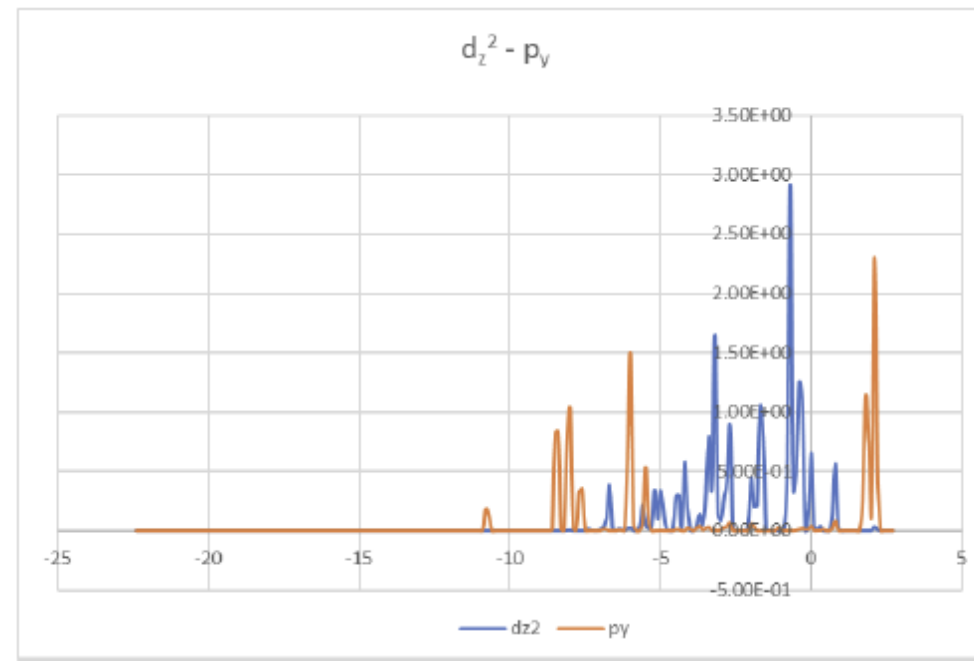
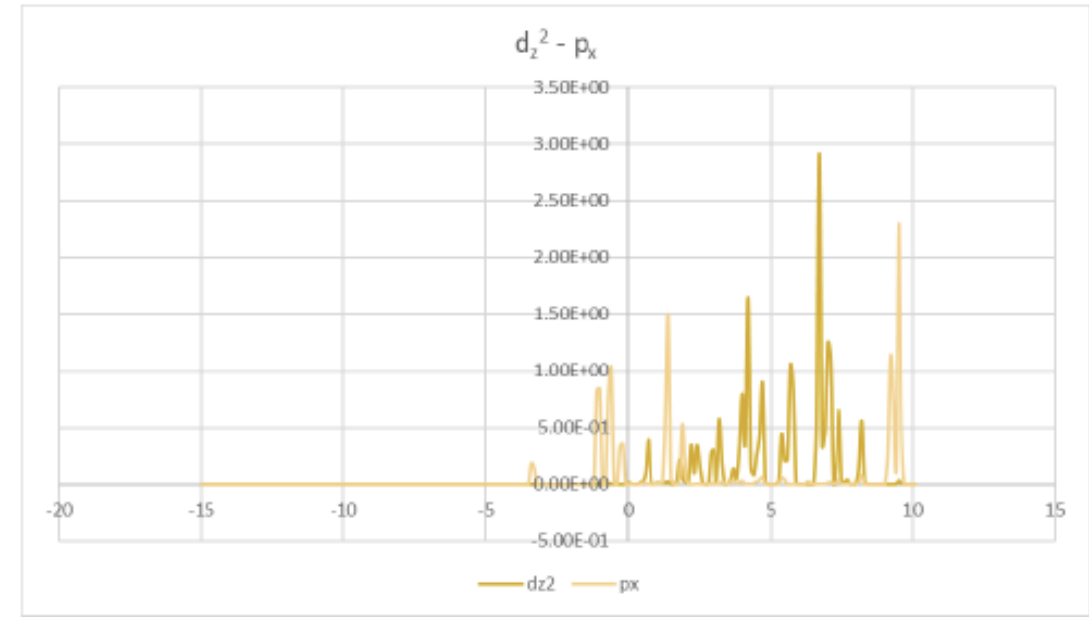
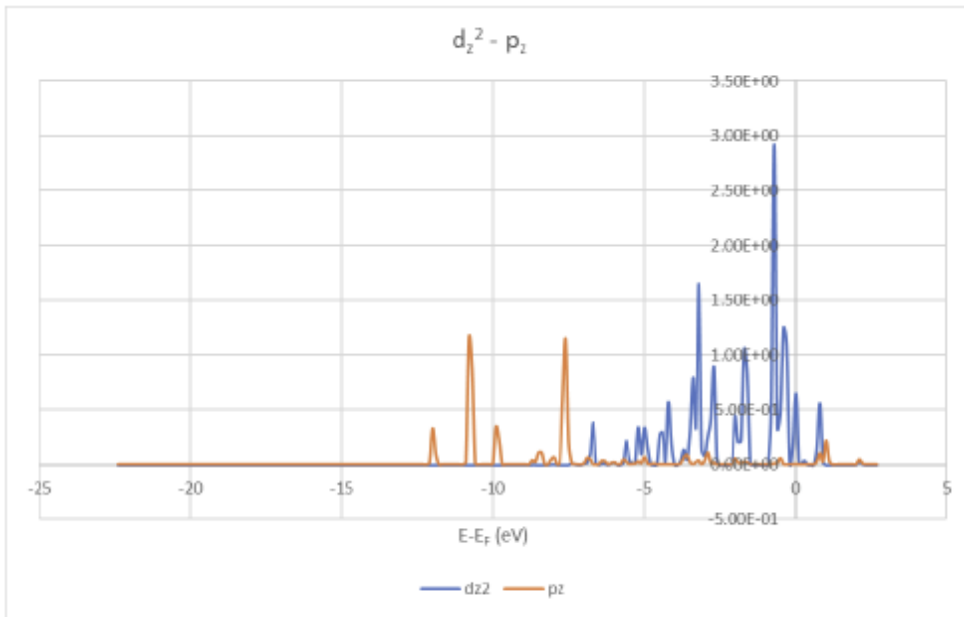
$$E_{ads} = -1.36 \text{ eV}$$

$$d_{C-O} = 1.157 \text{ \AA}$$

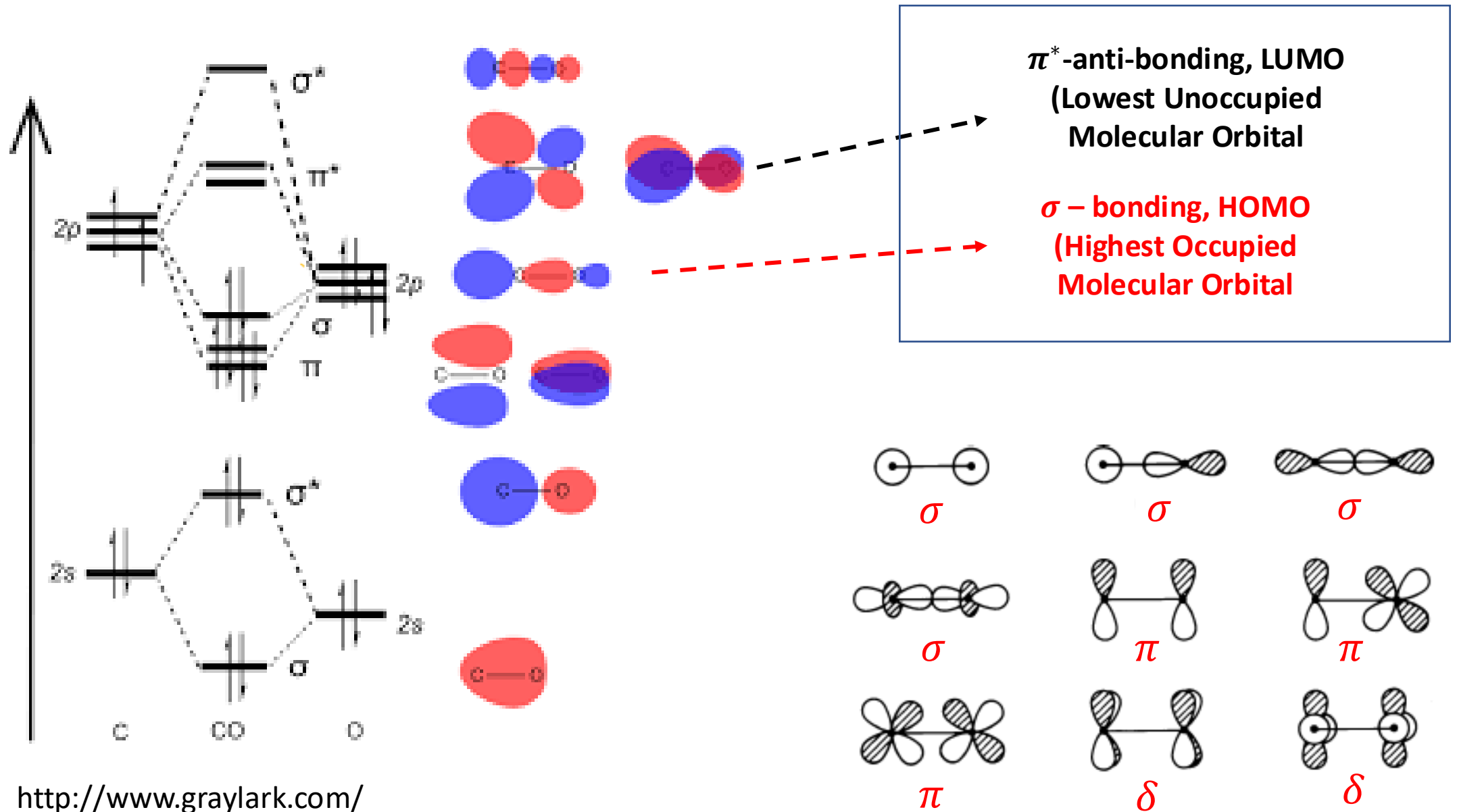
$$d_{C-Pt} = 1.837 \text{ \AA}$$

```
&DOS
  prefix = 'CO_Pt100_slab'
  outdir = './tmp'
  fildos = 'CO_Pt100_slab_dos'
  emin = -10.0
  emax = 10.0
/
```

```
&projwfc
  prefix      = 'CO_Pt100_slab'
  outdir      = './tmp'
  emin        = -10.0
  emax        = 10.0
  DeltaE      = 0.01
  filpdos     = 'CO_Pt100_slab_pdos'
/
```

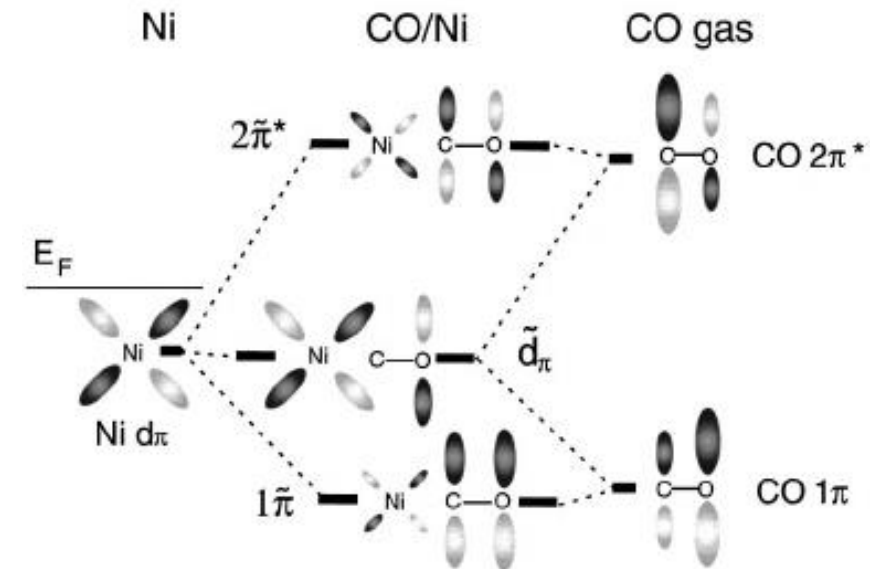
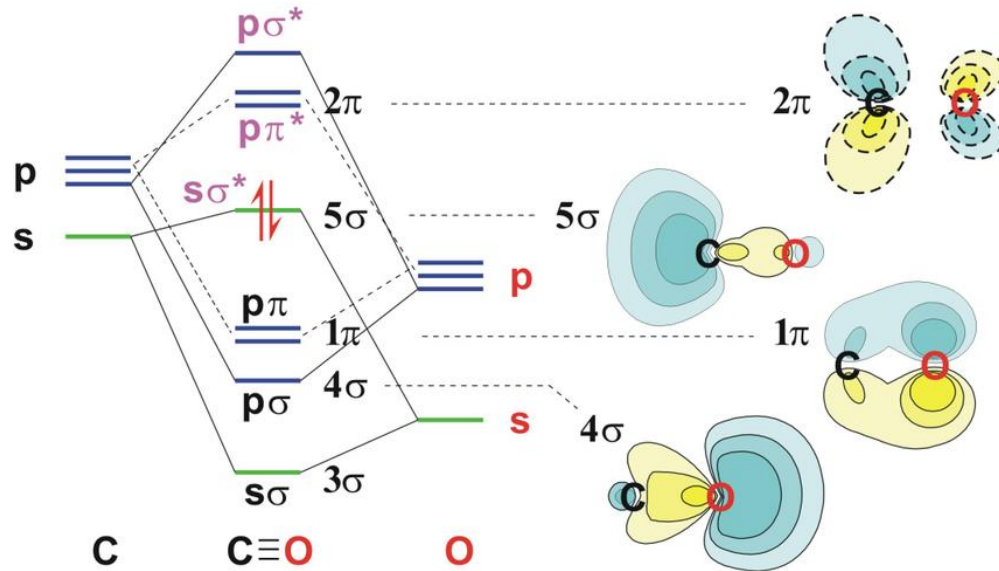
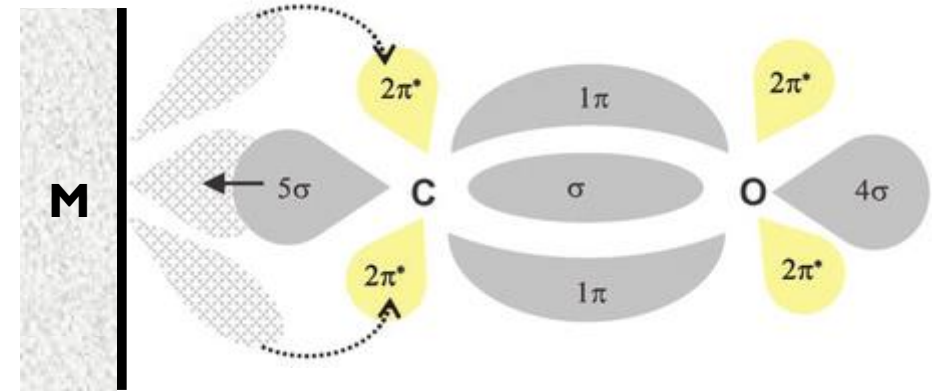
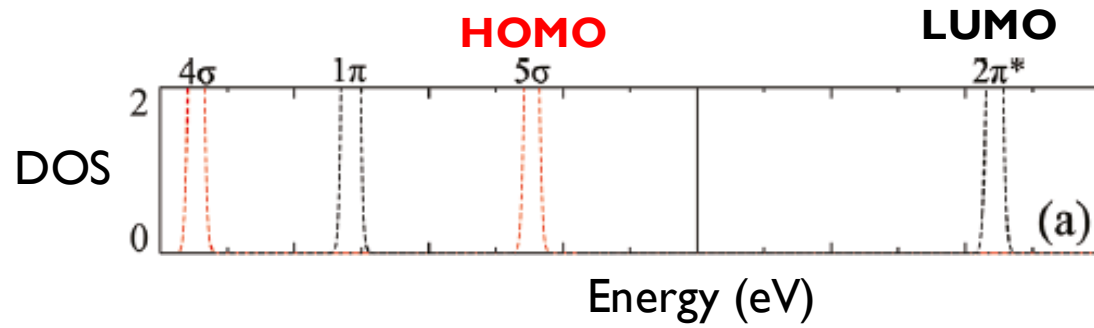


Molecular Orbital CO



Blyholder Mechanism

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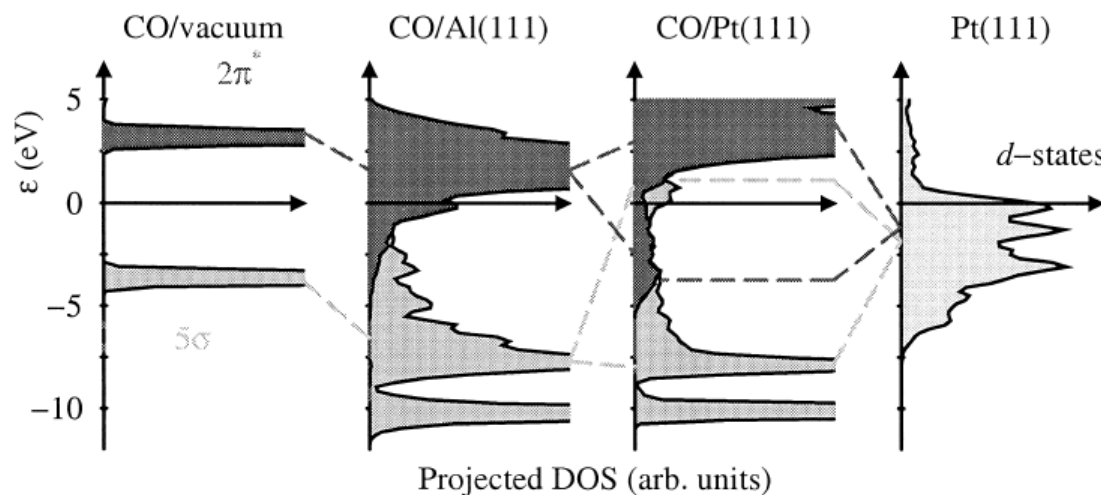
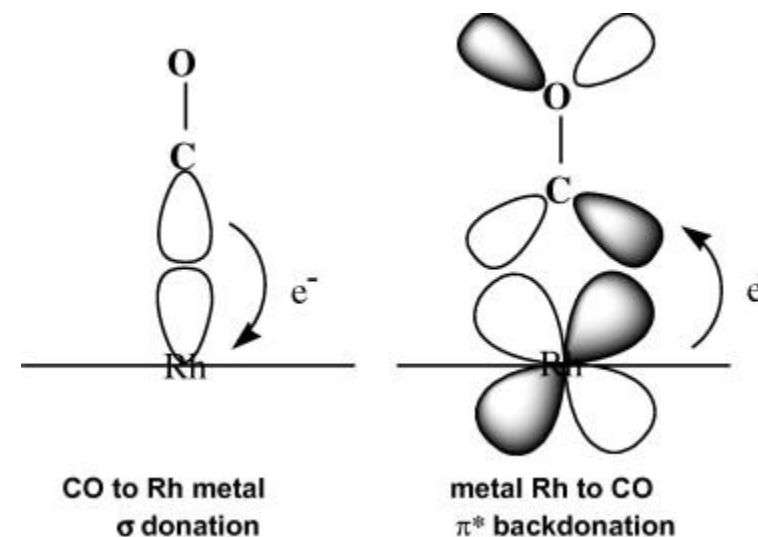


FIG. 12. The self-consistent electronic DOS projected onto the 5σ and $2\pi^*$ orbitals of CO: in vacuum and on Al(111) and Pt(111) surfaces. Also shown is the DOS from the d bands in the Pt(111) surface. The sharp states of CO in vacuum are seen to broaden into resonances and shift down in energy on the simple metal surface (mixing with the 4σ state causes additional structure in the 5σ resonance). On the transition metal surfaces the CO resonances further hybridize with the metal d states. This leads to shifts in the 5σ and $2\pi^*$ levels and to antibonding 5σ - d states at the top of the d bands and bonding $2\pi^*$ - d states at the bottom. These states have low weight in the 5σ and $2\pi^*$ projections shown. Adapted from Hammer *et al.* (55).



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